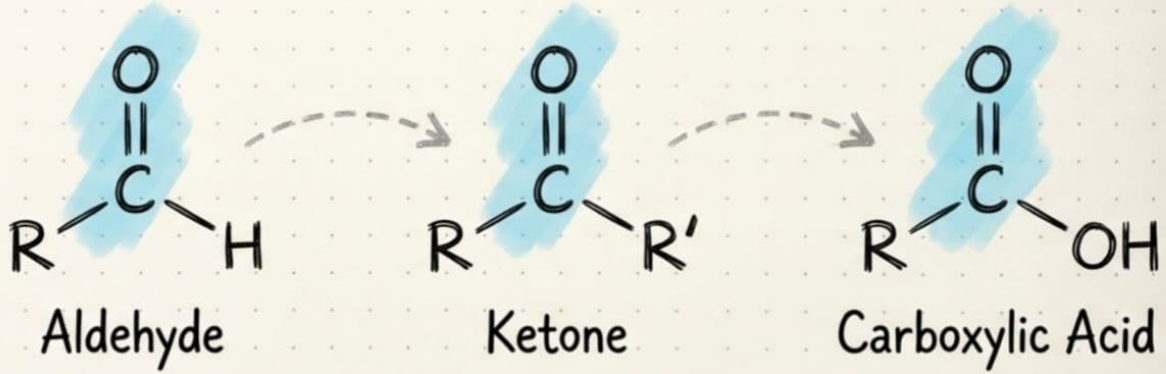
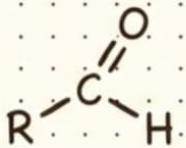
The Chemist's Playbook: Carbonyls & Carboxylic Acids

Synthesis, Behaviour, and Properties of Aldehydes, Ketones, and Carboxylic Acids



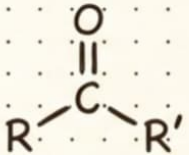
Section 1: The Carbonyl Twins

Aldehydes

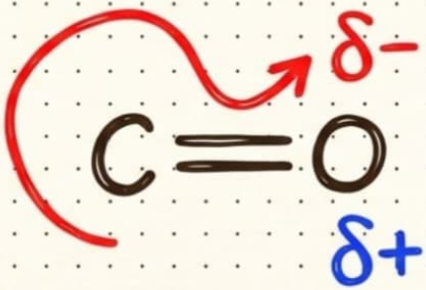


Terminal functional group.
Always at the end of the chain.

Ketones

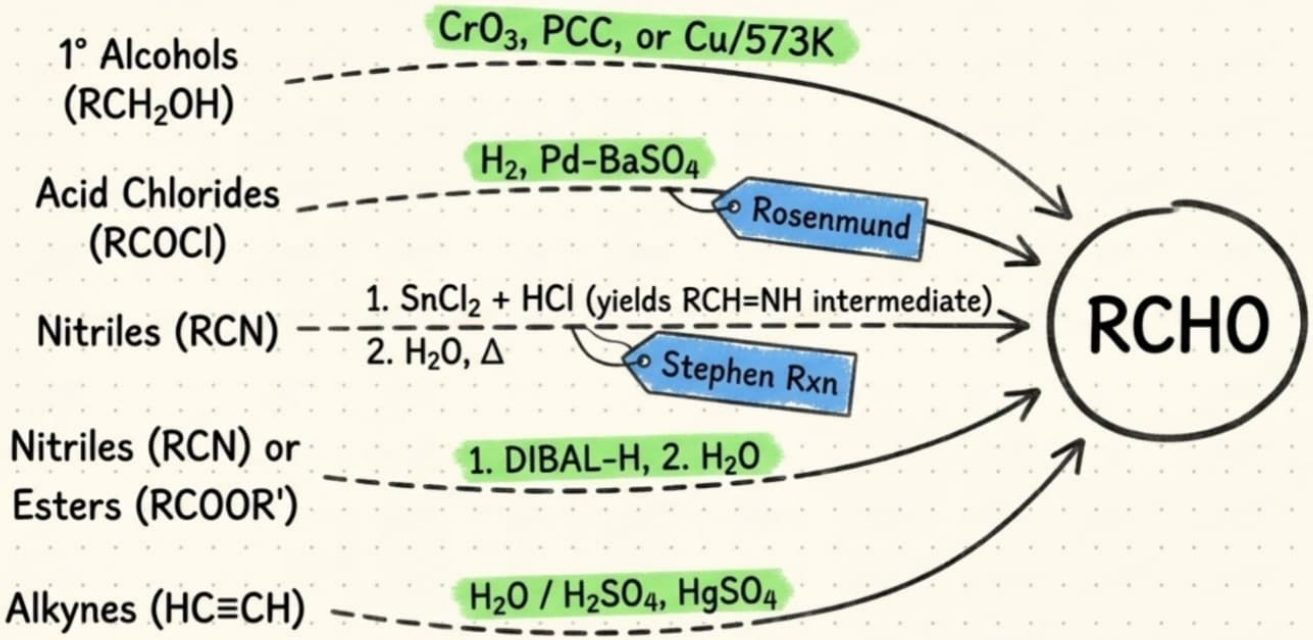


Internal functional group.
Flanked by carbon chains.

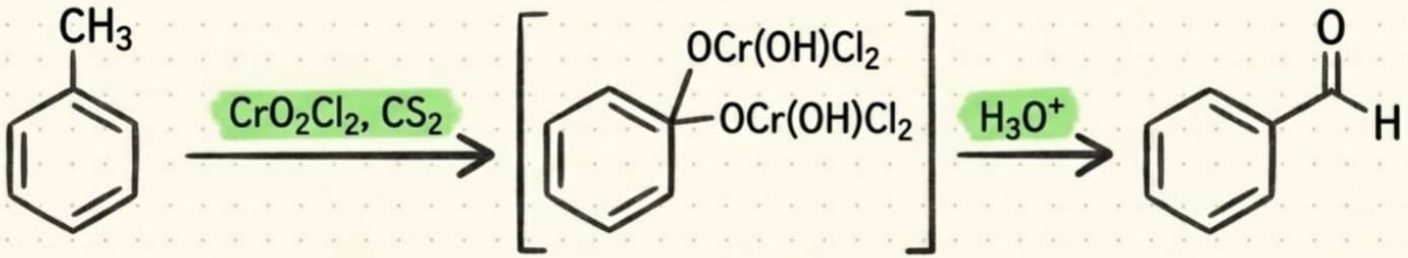


Both share the Carbonyl Group (>C=O). Their chemistry is dictated by the electrophilic (δ+) carbon!

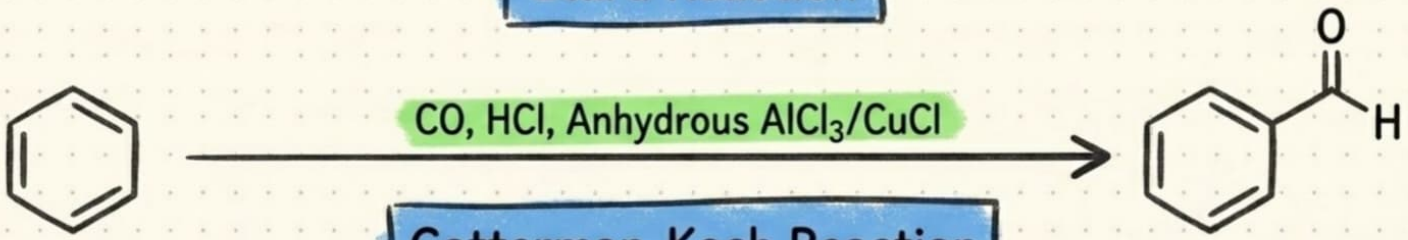
Synthesis Roadmap: Aliphatic Aldehydes (RCHO)



Synthesis Roadmap: Aromatic Aldehydes

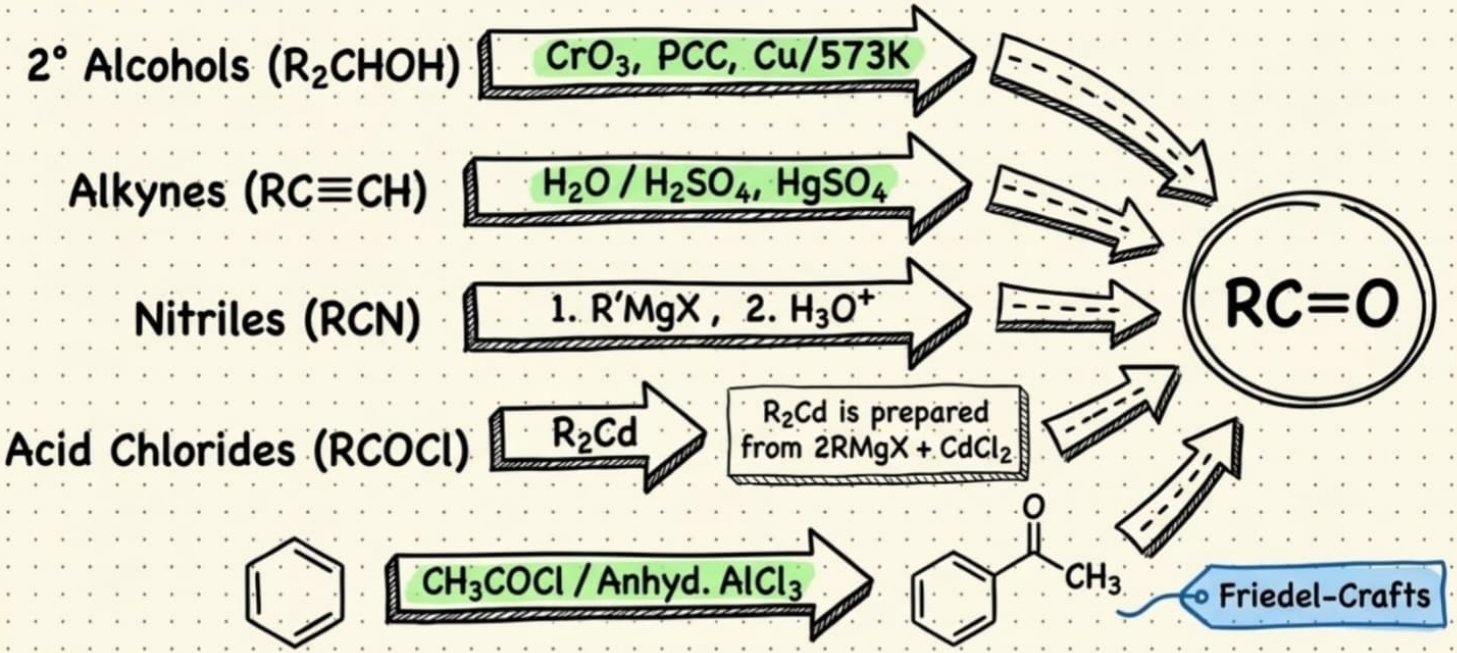


Etard Reaction



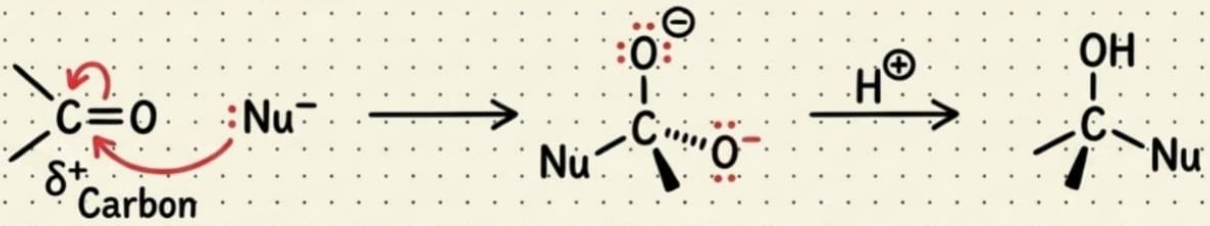
Gatterman-Koch Reaction

Synthesis Roadmap: Ketones (RCOR')

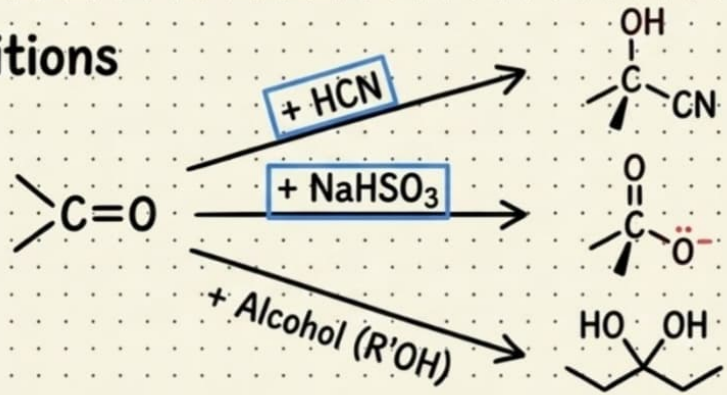


Behaviour: Nucleophilic Addition Hub

The Core Mechanism



Common Additions



Cyanohydrin (>C(OH)(CN))

Bisulfite addition compound

Hemiacetal → Acetal/Ketal

The Ammonia Derivatives Matrix (H_2N-Z)

$$>C=O + H_2N-Z \rightleftharpoons >C=N-Z + H_2O$$

Group "Z"	Reagent Name	Derivative Name (>C=N-Z)
$-H$	Ammonia (NH_3)	Imine
$-OH$	Hydroxylamine (NH_2OH)	Oxime
$-NH_2$	Hydrazine (NH_2NH_2)	Hydrazone
$-NH-Ph$	Phenylhydrazine	Phenylhydrazone
$-NH-C_6H_3(NO_2)_2$	2,4-DNP	2,4-dinitrophenylhydrazone
$-NH-CO-NH_2$	Semicarbazide	Semicarbazone

Reduction Routes: Back to Hydrocarbons

Carbonyl → ($LiAlH_4$ or $NaBH_4$ or H_2/Pd) → Alcohols (RCH_2OH or R_2CHOH)

Clemmensen Reduction

$$>C=O \xrightarrow{\text{Zn-Hg, HCl}} >CH_2 + H_2O$$

- Occurs under strongly acidic conditions.

Both entirely remove the oxygen, converting a carbonyl directly into a methylene ($-CH_2-$) group!

Wolff-Kishner Reduction

$$>C=O \xrightarrow{NH_2NH_2} \text{Hydrazone}$$

$$\rightarrow \text{KOH/ethylene glycol, } \Delta \longrightarrow >CH_2 + N_2$$

- Occurs under strongly basic conditions.

Oxidation & Qualitative Tests

The Rule: Aldehydes easily oxidise to Carboxylic Acids (RCOOH). Ketones require strong oxidants and undergo carbon-carbon bond cleavage.

Tollens' Test / Silver Mirror

$$RCHO + 2Ag(NH_3)_2^+ + 3OH^- \longrightarrow RCOO^- + 2Ag\downarrow + 4NH_3 + 2H_2O$$

Forms a brilliant silver mirror.

Fehling's Test

$$RCHO + 2Cu^{2+} + 5OH^- \longrightarrow RCOO^- + Cu_2O\downarrow + 3H_2O$$

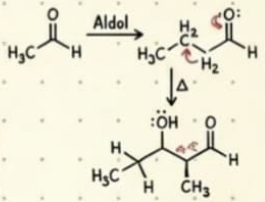
Yields a brick-red precipitate.

Haloform Reaction

$$\text{Methyl ketone } (R\text{-}CO\text{-}CH_3) \xrightarrow{NaOX} CHX_3 \text{ (Haloform)} + RCOONa$$

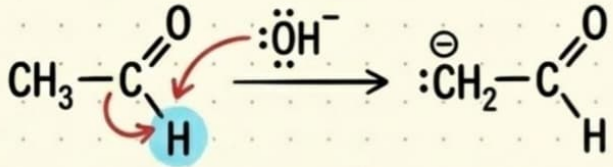
↳ Specific test for compounds with a terminal methyl ketone group.

The Aldol Condensation Masterclass

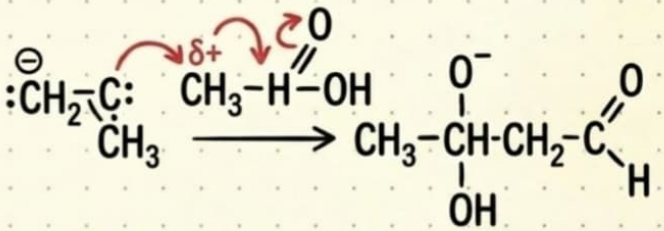


MUST have at least one α-Hydrogen! Reagent: dilute NaOH.

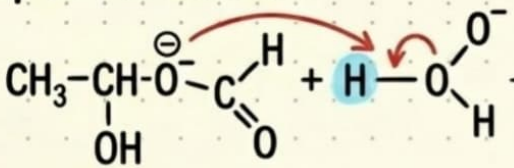
Step 1: Enolate Formation



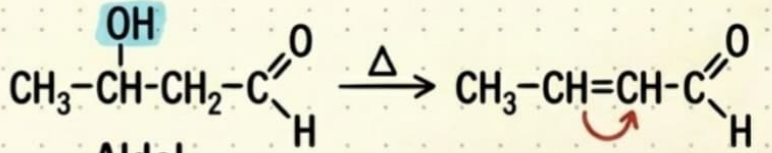
Step 2: Nucleophilic Attack



Step 3: Protonation



Step 4: Dehydration



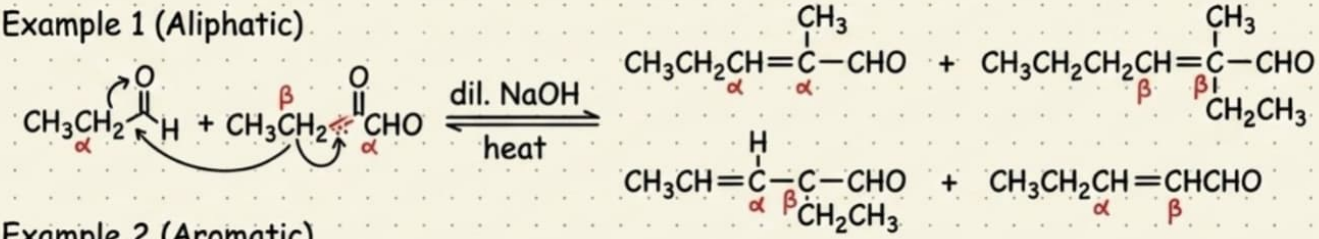
Aldol

Cross-Aldol Combinations

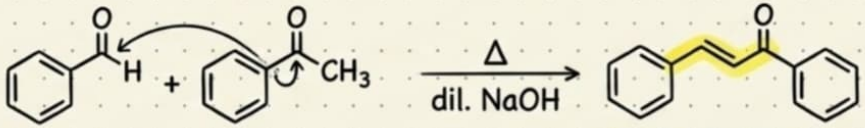
The Golden Rule

- OH group gets attached to the β-carbon.
- Methylene ($-CH_2-$) attacks the carbon attached to OH.
- The double bond always appears on the α-β carbon.

Example 1 (Aliphatic)



Example 2 (Aromatic)



Benzaldehyde Acetophenone Benzalacetophenone

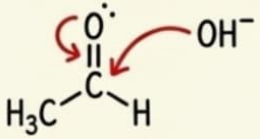
The Cannizzaro Reaction Masterclass

MUST HAVE NO α-Hydrogens!
Reagent: Concentrated KOH, Δ.

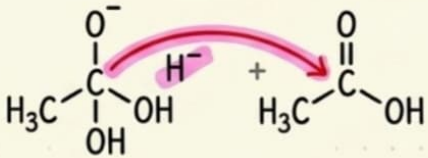
Core Concept: A disproportionation reaction (simultaneous self-oxidation and self-reduction).

General Reaction: $2\ HCHO \xrightarrow{\text{Conc. KOH, } \Delta} \underset{\text{(Alcohol)}}{CH_3OH} + \underset{\text{(Acid Salt)}}{HCOO^-K^+}$

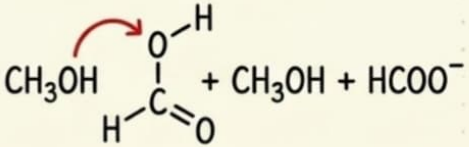
Step 1: Nucleophilic Attack



Step 2 (The Key Step): Hydride Transfer



Step 3: Rapid Proton Exchange



Physical Properties: Aldehydes & Ketones

State & Odour

Methanal is a gas.
Ethanal is a volatile liquid.
Others are liquids or solids.

Lower aldehydes possess a very pungent odour, becoming less pungent and more fragrant as the chain grows.

Boiling Points

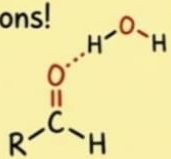
Higher BP than corresponding hydrocarbons due to weak intermolecular dipole-dipole attractions.

Lower BP than alcohols because they cannot form intermolecular H-bonds with themselves.

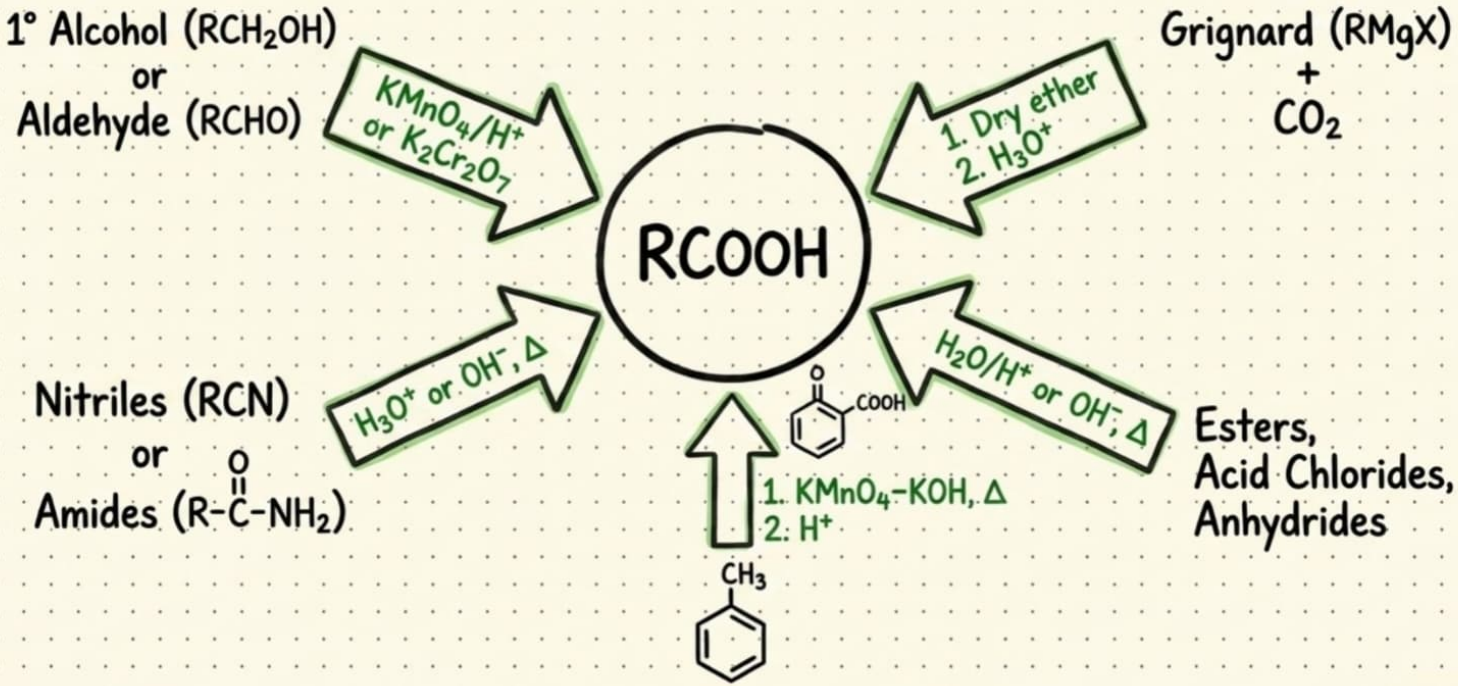
Solubility

Miscible in water up to 4 carbons!

Why? Because the carbonyl oxygen can form Hydrogen bonds with water molecules.



Section 2: Synthesis Roadmap for Carboxylic Acids



Reactions Dashboard: Bond Cleavages

Cleavage of O–H Bond → Acidity

$$RCOOH + Na \text{ or } NaOH \longrightarrow RCOONa \text{ (Salt)} + H_2/H_2O$$

$$RCOOH + Na_2CO_3 \longrightarrow RCOONa + H_2O + CO_2\uparrow$$

Effervescence of CO_2 is a classic test for carboxylic acids

Cleavage of C–OH Bond → Derivatives

$$RCOOH \xrightarrow{+\ PCl_5,\ PCl_3,\ \text{or}\ SOCl_2} RCOCl \text{ (Acid Chloride)}$$

$$RCOOH \xrightarrow{+\ R'OH,\ H^+} RCOOR' \text{ (Ester)}$$

$$RCOOH \xrightarrow{+\ NH_3,\ \Delta} RCONH_2 \text{ (Amide)}$$

$$RCOOH \xrightarrow[\text{dehydrating agents}]{\text{Heating with strong}} (RCO)_2O \text{ (Anhydride)}$$

Reactions Dashboard: Whole Group & α-Carbon

Reduction

$$RCOOH \xrightarrow[2.\ H_3O^+]{1.\ LiAlH_4 \text{ or } B_2H_6} RCH_2OH$$

Requires strong reducing agents. Will not stop at the aldehyde.

Decarboxylation

$$RCOONa \xrightarrow[\Delta]{NaOH \ \& \ CaO} RH + Na_2CO_3$$

Removes CO_2, stepping down the carbon chain length by 1.

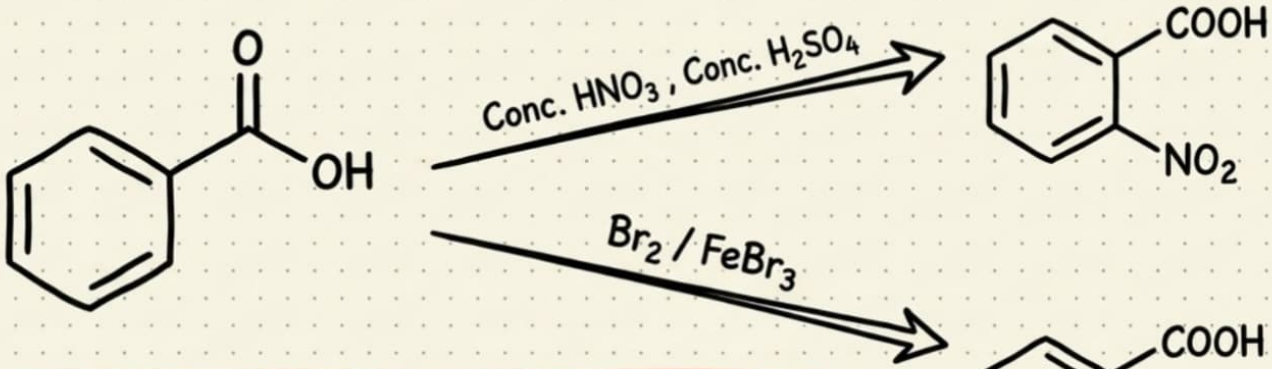
Hell-Volhard-Zelinsky

$$R\text{-}CH_2\text{-}COOH \xrightarrow[2.\ H_2O]{1.\ X_2 \ / \ Red\ P} R\text{-}\underset{\underset{X}{|}}{CH}\text{-}COOH$$

X = Cl, Br

Electrophilic Aromatic Substitution

The –COOH group is heavily electron-withdrawing. It deactivates the benzene ring and directs incoming electrophiles to the *meta* position.

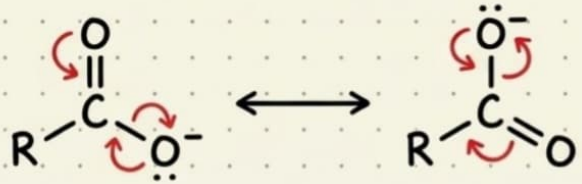


Friedel-Crafts reactions do NOT occur on benzoic acid! The carboxyl group deactivates the ring too much, and the catalyst ($AlCl_3$) gets bonded to the carboxyl group.

Acidity Deep Dive: Carboxylic Acids vs Phenols

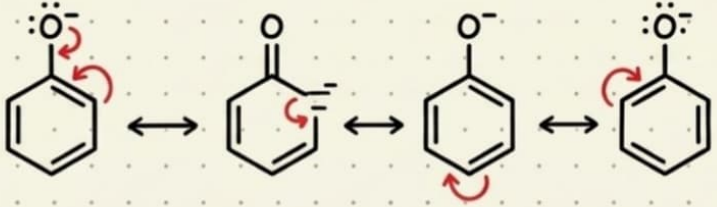
Carboxylic acids are more acidic than phenols because the carboxylate ion is substantially more stable than the phenoxide ion.

The Carboxylate Ion



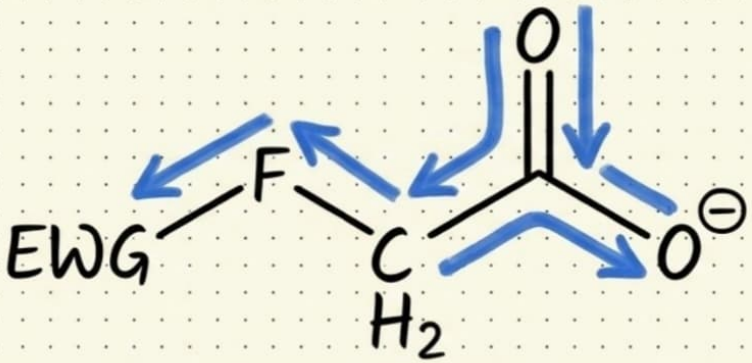
Negative charge is perfectly delocalised across TWO highly electronegative Oxygen atoms. Highly stable equivalent resonance structures.

The Phenoxide Ion



Negative charge is delocalised, but mostly onto less electronegative Carbon atoms in the ring. Less effective stabilisation.

Acidity Modifiers: The Inductive Effect

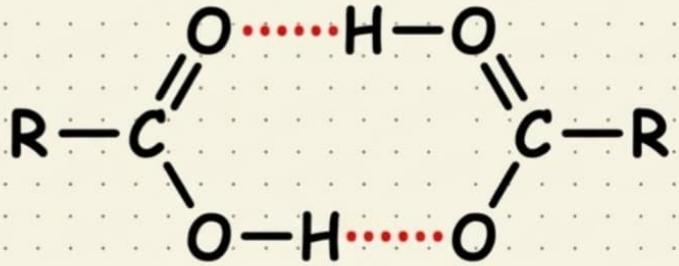


- Electron Withdrawing Groups (EWG) like halogens or $-NO_2$ pull electron density away via the -I effect.
- This disperses the negative charge on the carboxylate ion, stabilising the conjugate base, and therefore INCREASING acid strength.

Physical Properties: The Vapour Dimer

Boiling Point:
Significantly higher B.P. than corresponding alcohols, aldehydes, and ketones. The intermolecular Hydrogen bonding is exceptionally strong.

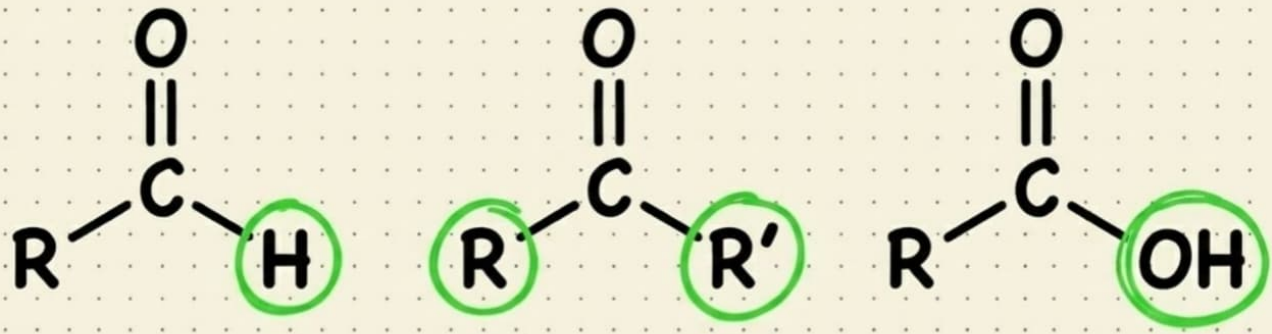
The Dimer Graphic



Hydrogen-bonded Dimer
(forms even in the vapour phase!)

Solubility: Soluble in water up to 4 carbon atoms. As the hydrophobic alkyl C-chain increases, solubility rapidly decreases. Benzoic acid is nearly insoluble in cold water.

The Carbonyl Thread



From nucleophilic **addition** to alpha-carbon chemistry, the behaviour of these molecules is entirely dictated by one common feature: the **polarisation of the Carbonyl Group**. Master the dipole, and you master the chemistry.